

Alessandro Porcheddu

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Education

University of Bologna

B.SC. IN COMPUTER SCIENCE AND ENGINEERING

Cesena, Italy

Sept. 2023 - Jul. 2026

- **Relevant Coursework:** Algorithms and Data Structures (**C**) - 4.0 GPA, Object-Oriented Programming (**Java**) - 4.0 GPA, Linear Algebra for Machine Learning (**Python**) - 4.0 GPA, Programming (**C**) - 4.0 GPA
- **Extracurricular Activities:** Member of the **E-Powertrain and Electronics Department** of the university's motorsport team (**MotoStudent Project**), contributing to the development of an electric motorcycle.

I.T.T. Montani

HIGH SCHOOL DIPLOMA

Fermo, Italy

Sept. 2018 - Jun. 2023

- Designed and implemented automation systems involving **PLC** and **HMI** programming.
- Participated in the **Omron Smart Project Trophy 2023** competition.
- Earned a Omron **Robotics Operation and Programming Certificate**.

Work Experience

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SYSTEM ADMINISTRATOR INTERN

Cesena, Italy

Sept. 2025 - Ongoing

- Assisted in configuring and maintaining Windows **Domain Controller**, ensuring reliable system performance and uptime.
- Documented system configurations, network layouts, and **maintenance procedures** for internal use.
- Worked with virtualization tools such as **VMware and Proxmox** to create and test system environments.

Cables

AUTOMATION TECHNICIAN

Cesena, Italy

Jun. 2023 - Jun. 2023

- Assisted in assembling electrical control panels for **industrial automation systems**, following detailed wiring diagrams and specifications.
- Helped troubleshoot basic PLC input/output issues and performed routine checks on control system hardware.

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IT ASSISTANT

Wrocław, Poland

Jun. 2022 - Jul. 2022

- Completed the **Erasmus+** "Trainee for Industry 4.0 Plus" internship program, gaining practical experience in industrial environments.
- Assisted with the digitization of paper records by transferring them to solid-state storage systems.
- Participated in basic electrical work including the installation of sockets and lighting.

Side Projects

JavaDyno (Engine Dynamometer Management Software)

 GitHub

SOFTWARE DEVELOPMENT

- Developed an engine dynamometer management system (**EDMS**) with a group project team, overseeing complete data retrieval and graphical plotting.
- Implemented an **MVC** core structure, enabling data exportation and session logging capabilities.
- Ensured **OBD-II** over **CAN** compatibility for in-vehicle testing, enhancing real-world applicability.
- Integrated **WebSocket** and **Serial** connection support for versatile device communication and included simulation features for comprehensive testing.

Motorcycle Parts E-commerce Platform

Website

WEB DEVELOPMENT

- Built a full-stack e-commerce platform using **Laravel 12**, **React 19**, and **Inertia.js**.
- Implemented secure payment processing with **Stripe** via **Laravel Cashier**.
- Designed a responsive, mobile-first UI with reusable components using the **shadcn** React library.
- Developed an **admin dashboard** for product, order, and customer management with analytics and reporting tools.
- Ensured robust security with **mandatory email verification**, protected routes, and **Stripe webhook signature validation**.

Motorcycle Exhaust Valve Control Unit

EMBEDDED SYSTEMS

- Developed an electronic exhaust valve control unit based on **ESP32**.
- Created a browser app to control the device via WiFi through **WebSocket** communication.
- Designed a custom PCB using **KiCad** and a custom 3D printed enclosure using **Fusion360**.
- Implemented **secure OTA** updates through a custom remote server based on NodeJS.